

Multi-Region EC2 Traffic Routing with AWS Global Accelerator

To Begin with the Lab

Summary of the Lab

In this lab, **AWS Global Accelerator** was used to route user traffic between two **Amazon EC2** instances deployed in different regions, Mumbai and N. Virginia. Two web servers were launched with region-specific user data so each instance served different content based on its location. A Global Accelerator was then created with two endpoint groups, one for each region, with traffic dials set to balance requests.

Health checks were configured to monitor the endpoints and keep traffic flowing to healthy instances. Finally, the accelerator's static IPs were used to verify that users were directed to the correct regional server, demonstrating global traffic routing and improved availability.

Understanding AWS Global Accelerator

AWS Global Accelerator improves application availability and performance by routing traffic through the AWS global network instead of the public internet. It gives applications static IP addresses and automatically sends users to the nearest healthy endpoint, which reduces latency and improves response time. It also supports health checks and failover so traffic can move away from an unhealthy region without changing the application endpoint. This makes it a good fit for applications that serve users across multiple regions.

Example:

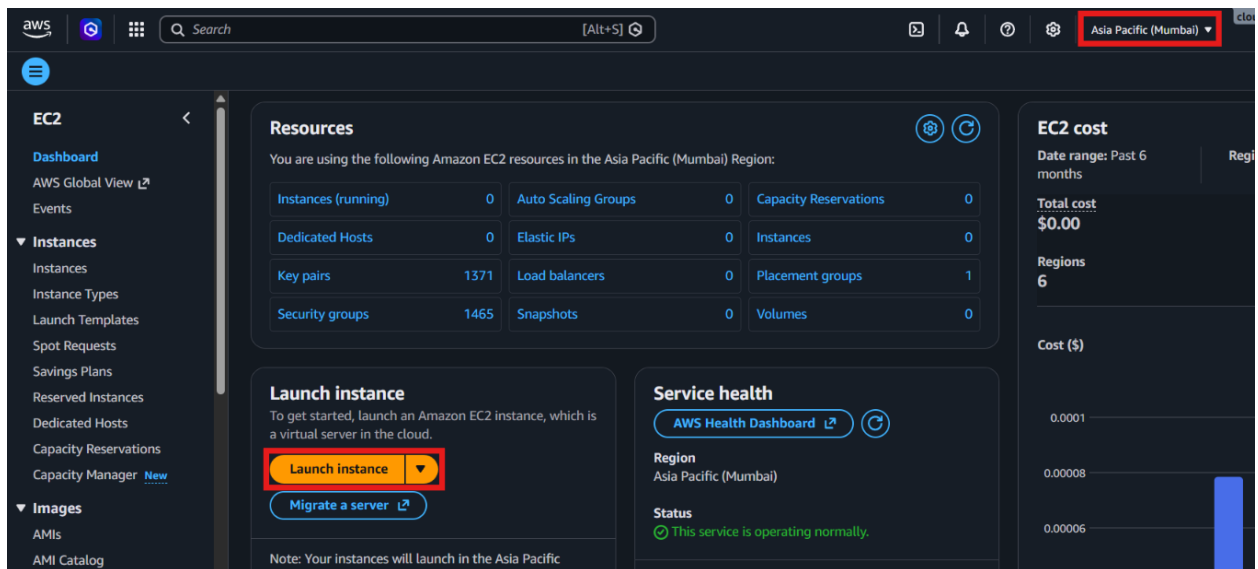
A website serves Indian users from a Mumbai EC2 instance and US users from a N. Virginia EC2 instance. When the Mumbai server is healthy, users in India are routed there, while users in the United States are routed to the N. Virginia server. If one region becomes unavailable, **AWS Global Accelerator** can redirect traffic to the healthy region, keeping the application accessible.

Lab Steps

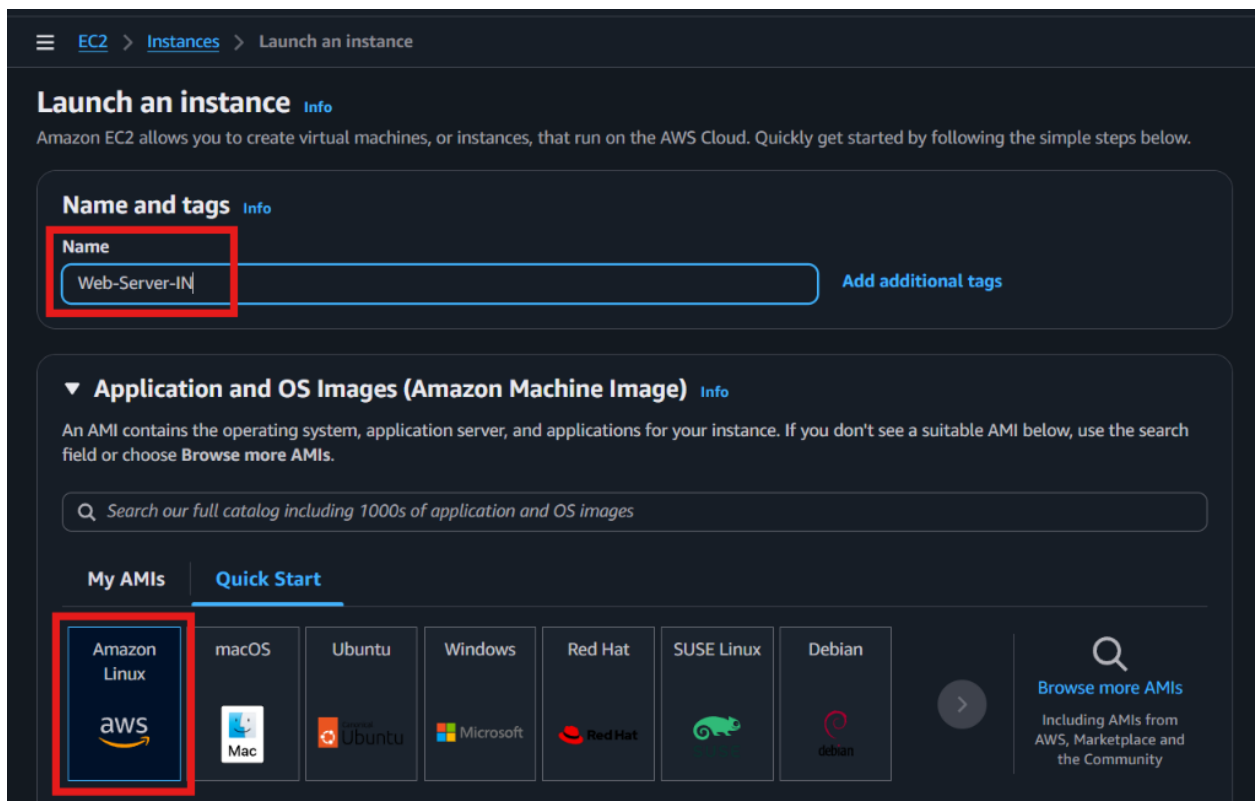
- Sign in to the AWS Management Console.

1. Create Instances for Respective Regions

- Launch an **Amazon EC2** instance in the Mumbai region (**ap-south-1**).



- Enter the Proper name for identifying the Instance for the region, in my case I have taken Web-Server-IN.
- Choose the **Amazon Linux 2023** AMI.



- Select the **t3.micro** instance type.
- Select **Key pair** for this or create one by yourself.

▼ Instance type Info | Get advice

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

All generations
Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

June-2026

Create new key pair

▼ Network settings Info

Network Info

vpc-072c4acd761c3b942

Edit

- Click **Allow HTTP traffic from internet** and make sure you have enabled public IP.
- Open port 80 (HTTP) in the security group.
- Click **Advanced details** and at the very bottom you will find **User data**.
- Configure user data for the Mumbai web server and click **Launch instance**.

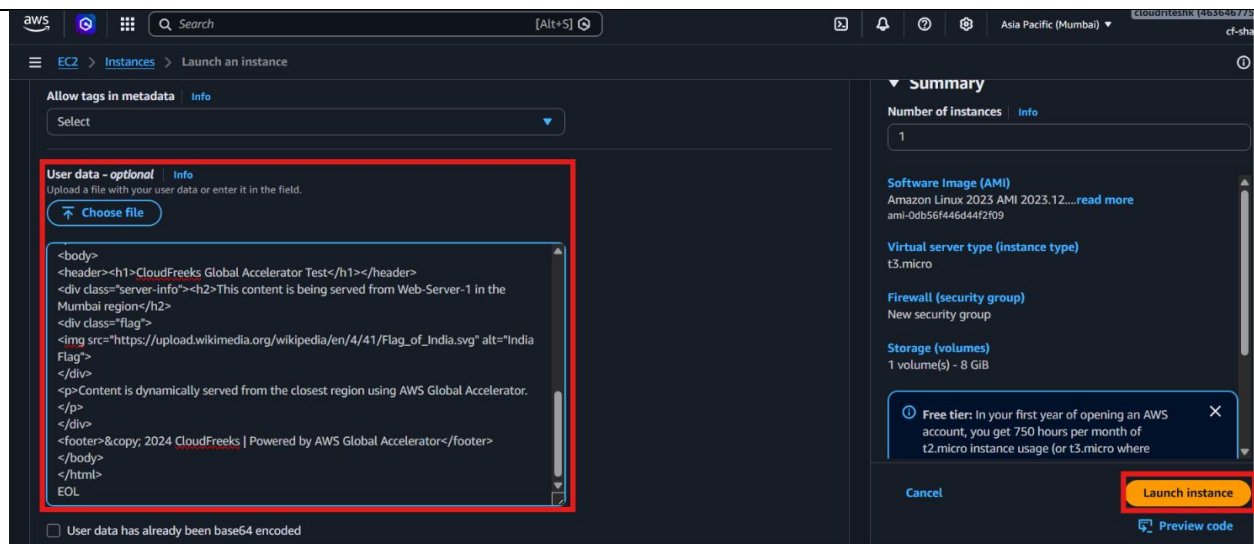
```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
cat <<EOL > /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>CloudFreeeks Global Accelerator Demo</title>
<style>
body { font-family: Arial, sans-serif; text-align: center; background-color: #f5f5f5; }
header { background-color: #1e73be; padding: 20px; }
h1 { color: white; }
.server-info { margin: 50px auto; padding: 20px; border: 1px solid #ddd; background-color: #fff;
max-width: 600px; }
.server-info h2 { margin: 0; padding: 10px 0; }
.flag img { width: 100px; margin: 20px 0; }
```

```

footer { background-color: #1e73be; padding: 10px; color: white; position: fixed; bottom: 0;
width: 100%; }
</style>
</head>
<body>
<header><h1>CloudFreeks Global Accelerator Test</h1></header>
<div class="server-info"><h2>This content is being served from Web-Server-1 in the Mumbai
region</h2>
<div class="flag">

</div>
<p>Content is dynamically served from the closest region using AWS Global Accelerator.</p>
</div>
<footer>&copy; 2024 CloudFreeks | Powered by AWS Global Accelerator</footer>
</body>
</html>
EOL

```



- Launch another **Amazon EC2** instance in the N. Virginia region (**us-east-1**).
- Enter the Proper name for identifying the Instance for the region, in my case I have taken Web-Server-USA.
- Choose the **Amazon Linux 2023 AMI**.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name: Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS Images

My AMIs Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0152204c1a187337c (64-bit (x86_64), uefi-preferred) / ami-02c114835b4e7739f (64-bit (ARM), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Summary

Number of instances: 1 Info

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-0152204c1a187337c

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) used with free tier AMIs, 750 hours per month of IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included.

Cancel

- Select the **t3.micro** instance type.
- Select **Key pair** for this or create one by yourself.

Instance type Info | Get advice

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

June-2026-us Create new key pair

Network settings Info

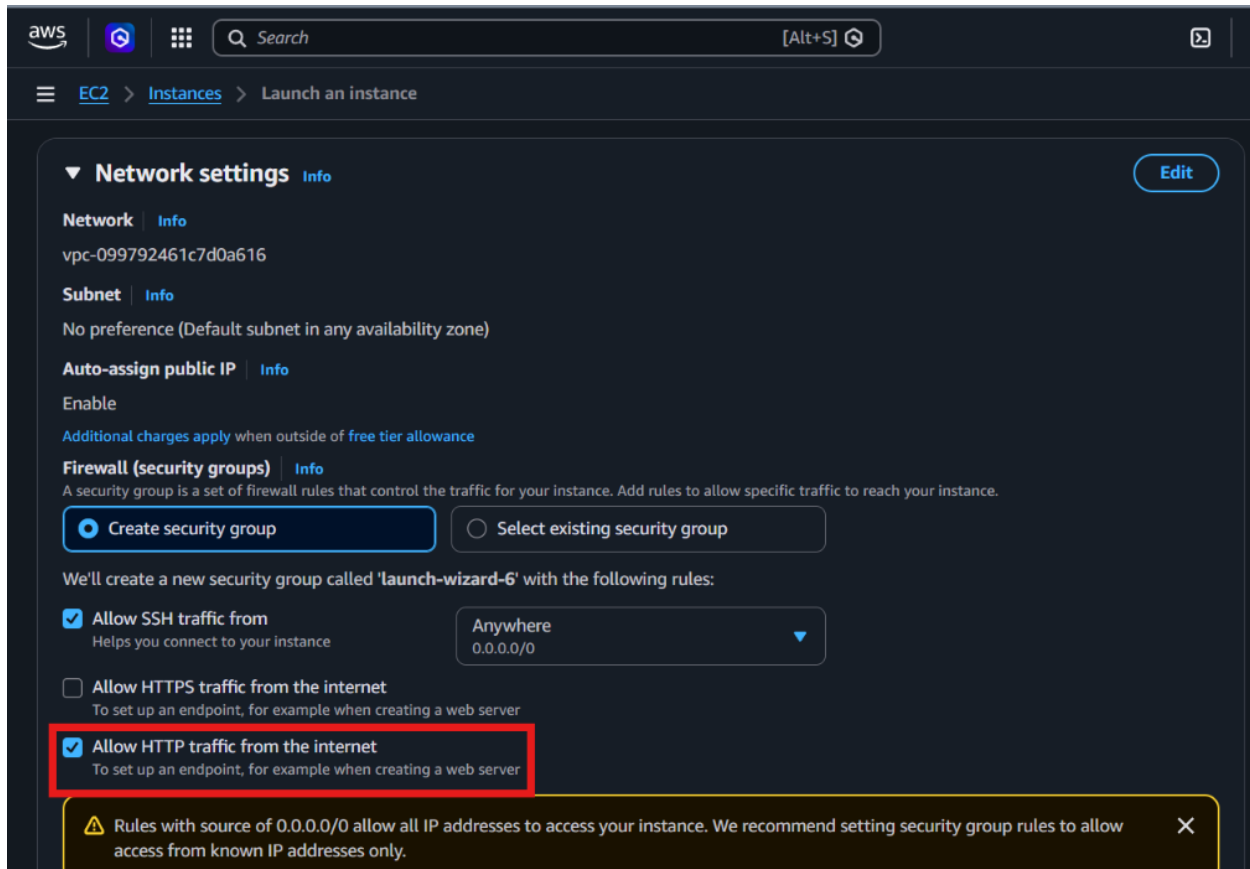
Network Info

vpc-099792461c7d0a616

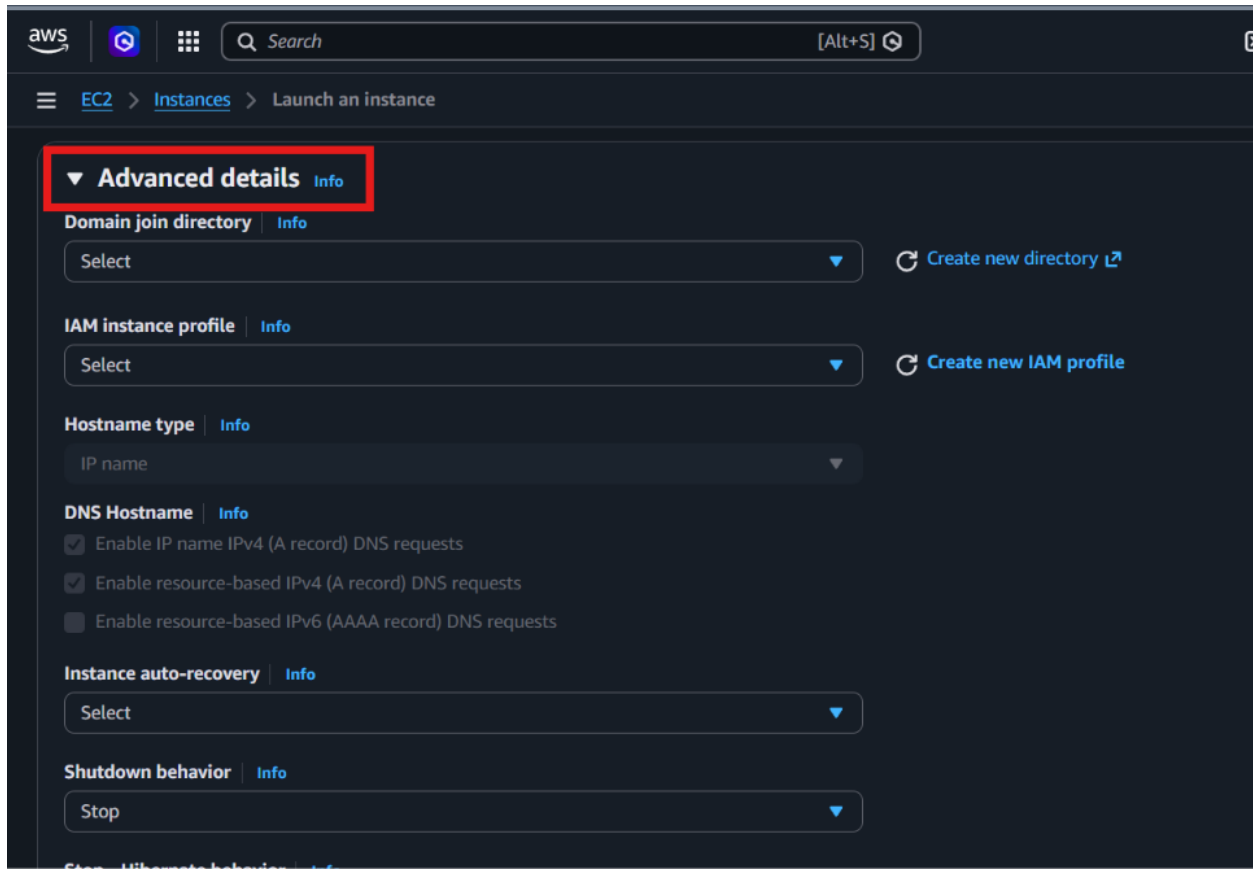
Subnet Info

Edit

- Click **Allow HTTP traffic from internet** and make sure you have enabled public IP.



- Open port 80 (**HTTP**) in the security group.
- Click **Advanced details** and at the very bottom you will find **User data**.



- Configure user data for the N. Virginia web server and click **Launch instance**.

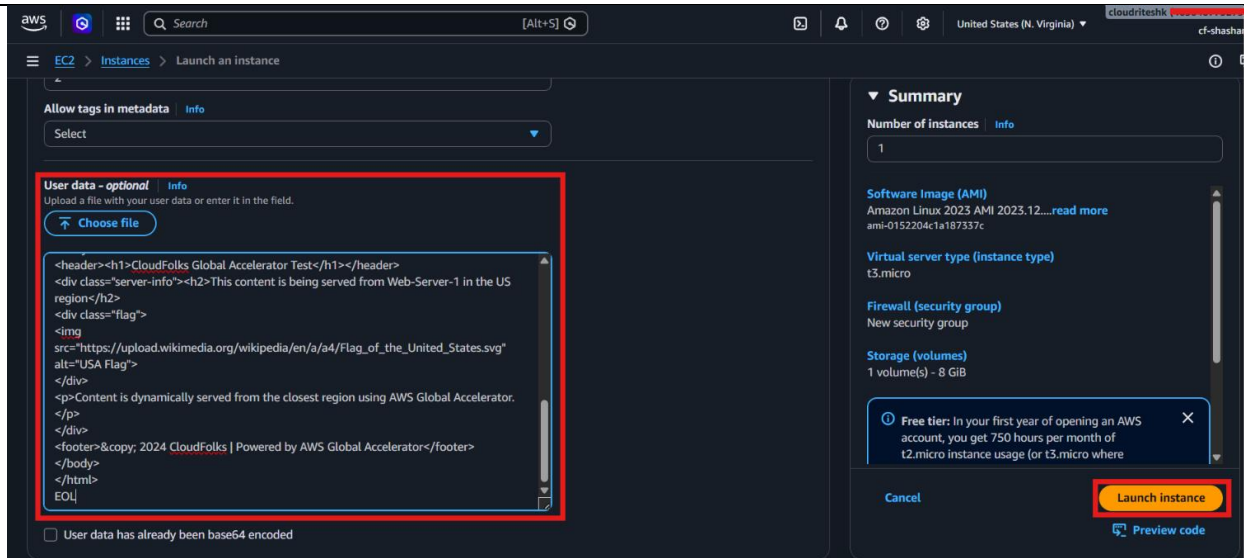
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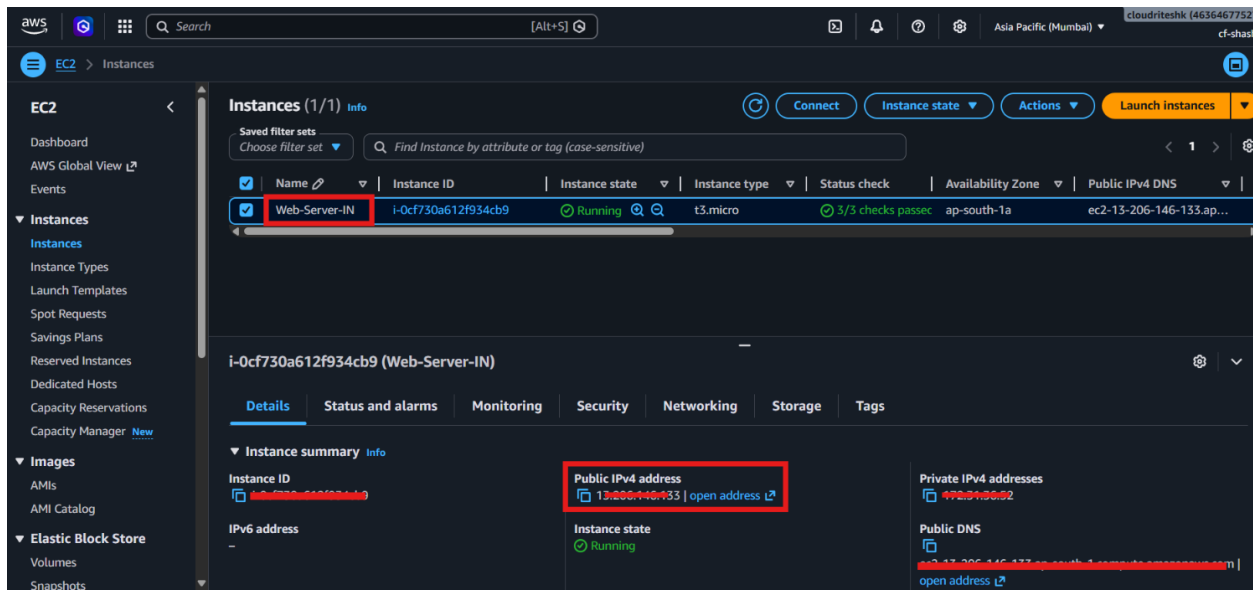
.flag img { width: 100px; margin: 20px 0; }
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</style>
</head>
<body>
<header><h1>CloudFolks Global Accelerator Test</h1></header>
<div class="server-info"><h2>This content is being served from Web-Server-1 in the US
region</h2>
<div class="flag">

</div>
<p>Content is dynamically served from the closest region using AWS Global Accelerator.</p>
</div>
<footer>&copy; 2024 CloudFolks | Powered by AWS Global Accelerator</footer>
</body>
</html>
EOL

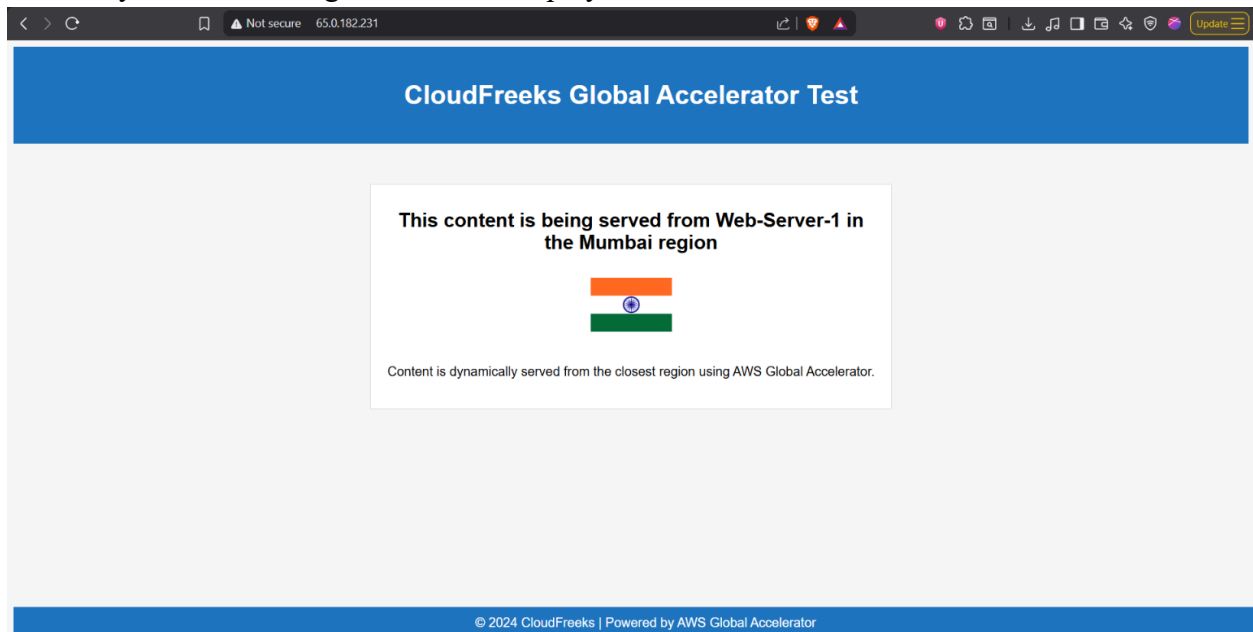
```



- Copy the public IP address of each instance.

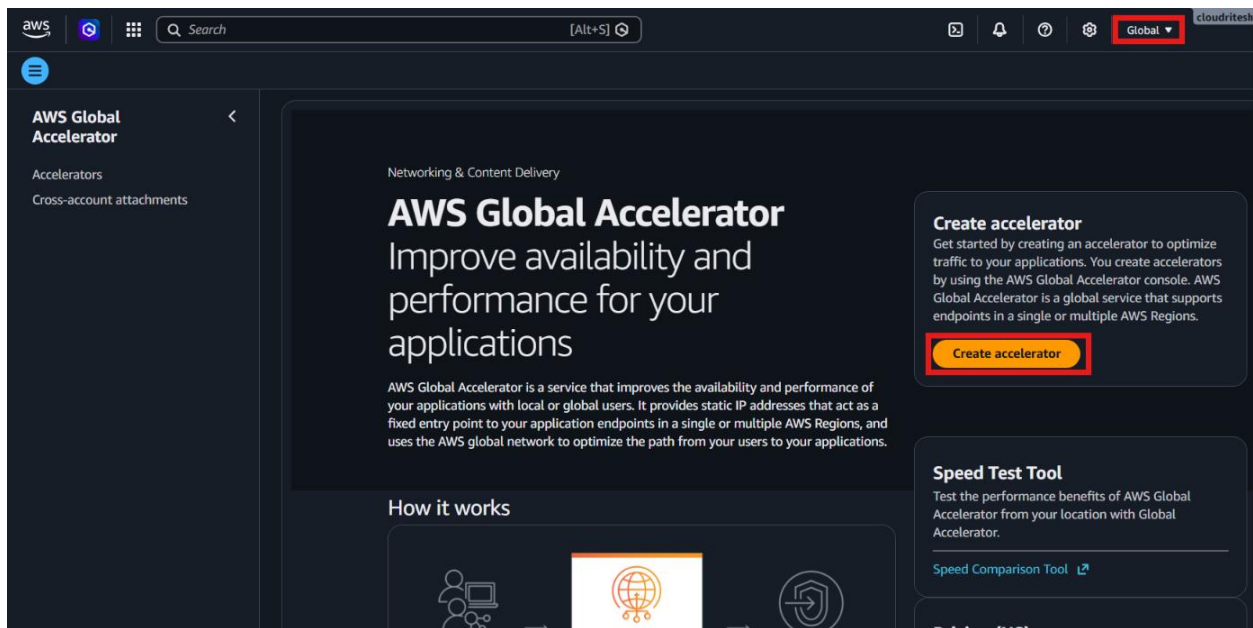


- Open both public IPs in a browser.
- Verify that the Mumbai instance displays Indian content.
- Verify that the N. Virginia instance displays USA content.

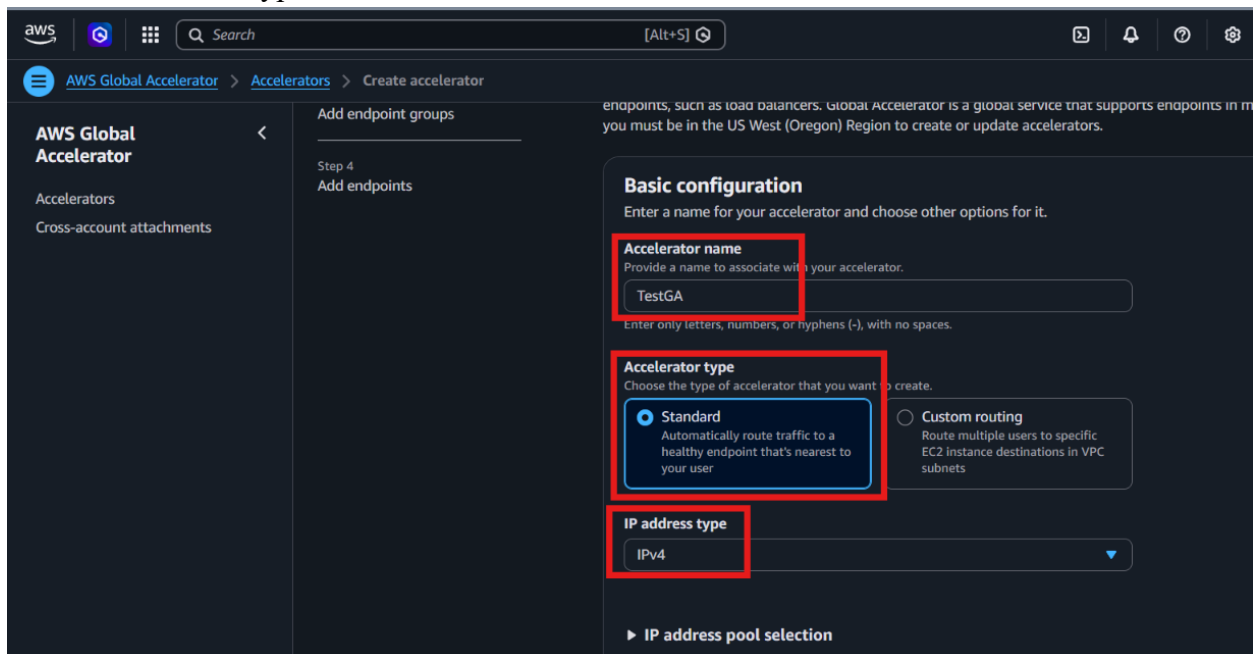


2. Configure AWS Global Accelerator

- Open **AWS Global Accelerator** in the AWS Management Console.
- Click **Create Accelerator**.



- Enter a name for the accelerator.
- Keep accelerator type as **Standard**.
- Use IP address type as **IPv4** for now and click **Next**.



- Add a listener with protocol **TCP**.
- Set the listener port to **80**.

Accelerators > Create accelerator

Step 1
Enter name

Step 2
Add listeners

Step 3
Add endpoint groups

Step 4
Add endpoints

Add listeners

A listener is a process that checks for connection requests that arrive to an assigned set of static IP addresses on a port or port range that you specify.

Listeners

You designate a listener by choosing a specific port or port range to listen on.

Listener 1

Ports ⓘ
80
Use commas to separate port numbers or ranges.

Protocol ⓘ
TCP

Client affinity ⓘ
None

Remove

Add listener

Cancel Previous **Next**

- Add an endpoint group for **N. Virginia (us-east-1)**.
- Select the N. Virginia EC2 instance as the endpoint.
- Set the traffic dial to **100**.
- Add an endpoint group for **Mumbai (ap-south-1)**.
- Select the Mumbai EC2 instance as the endpoint.
- Set the traffic dial to **100**.

Accelerators > Create accelerator

Step 1
Enter name

Step 2
Add listeners

Step 3
Add endpoint groups

Step 4
Add endpoints

Add endpoint groups

An accelerator includes one or more listeners that direct traffic to one or more endpoint groups. An endpoint group includes endpoints, such as load balancers.

Listener: 80 TCP

Each listener can have multiple endpoint groups. Each endpoint group can only include endpoints that are in one Region. You aren't required to add an endpoint group, but until you do, traffic to this listener won't reach any endpoints.

Endpoint group 1

Region ⓘ
us-east-1

Traffic dial ⓘ
100
A number from 0 to 100.

Remove

► Configure port overrides

► Configure health checks

Endpoint group 2

Region ⓘ
ap-south-1

Traffic dial ⓘ
100
A number from 0 to 100.

Remove

► Configure port overrides

► Configure health checks

Add endpoint group

Cancel Previous **Next**

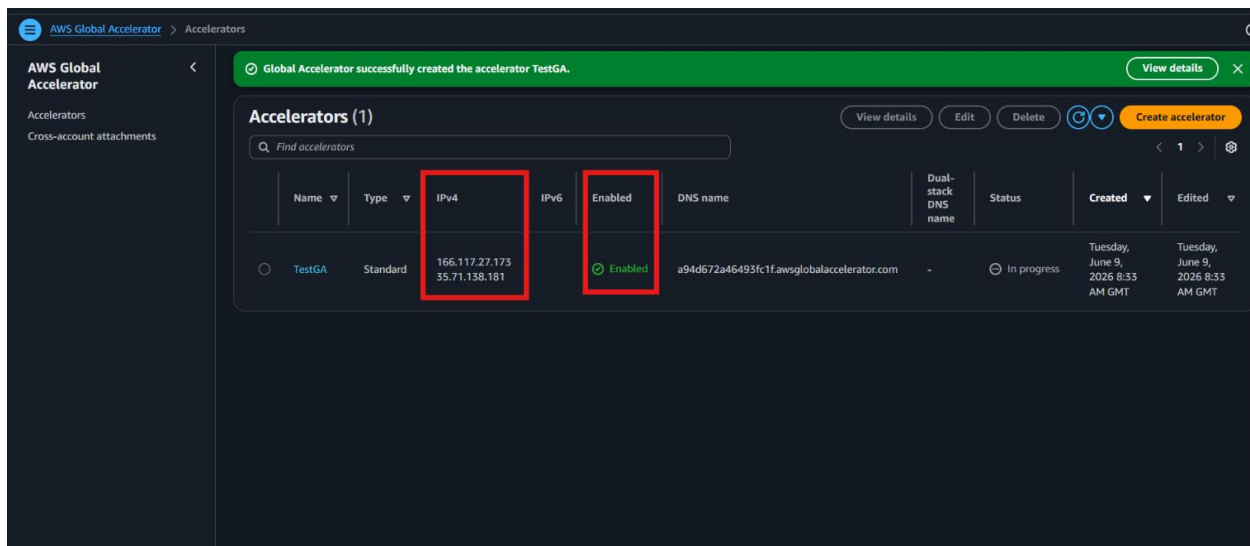
Configure Endpoint Groups

- Under **Endpoint type**, choose **Amazon EC2 instances**.
- Select the N. Virginia EC2 instance that was created earlier.

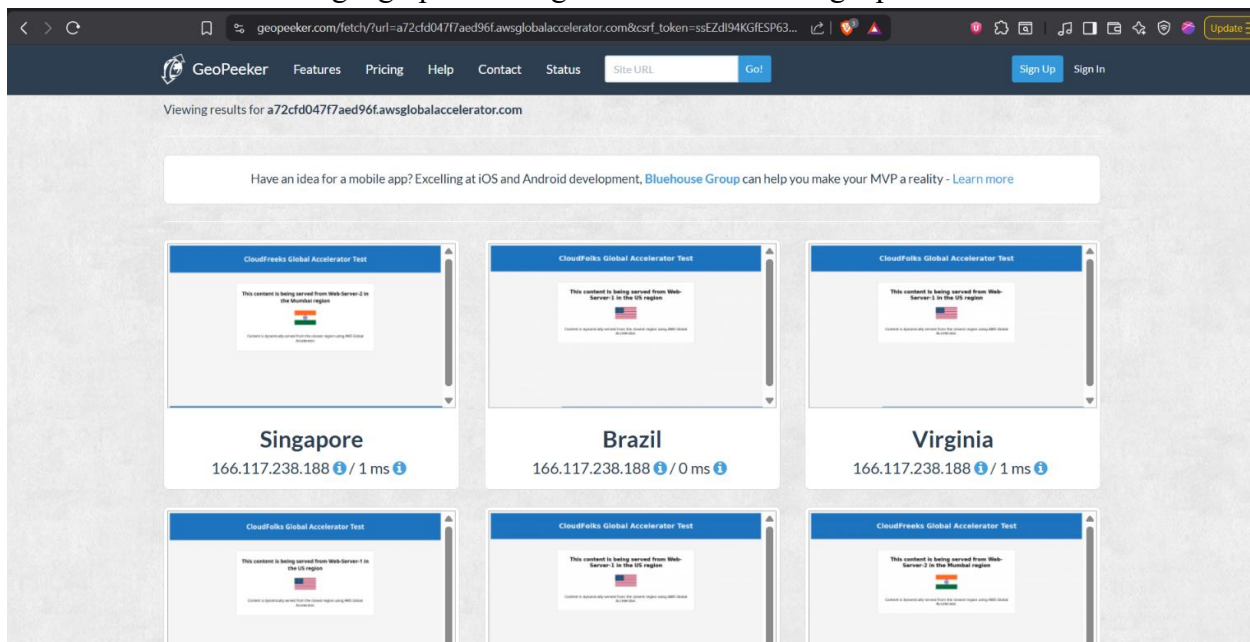
- Leave **Endpoint weight** at the default value.

- Select **Region** → **Asia Pacific (Mumbai) - ap-south-1**.
- Under **Endpoint type**, choose **Amazon EC2 instances**.
- Select the Mumbai EC2 instance that was created earlier.
- Leave **Endpoint weight** at the default value.

- Verify that both endpoint groups appear:
 - **N. Virginia (us-east-1)** → N. Virginia EC2 instance
 - **Mumbai (ap-south-1)** → Mumbai EC2 instance
- Leave all remaining settings as default.
- Click **Next** and proceed to **Review and Create Accelerator**.
- Review and create the accelerator.
- Use the provided static IPs to access the application.



- Verify that traffic from India is routed to the Mumbai instance and traffic from the USA is routed to the N. Virginia instance.
- For this we can use a geographic testing tool or website like [geopeeker.com](https://www.geopeeker.com)



- Confirm that **AWS Global Accelerator** successfully routed traffic between the two regional **Amazon EC2** instances.